

Group projects Programming in Python II

Deadline: 14/12/2024, 23:59, please submit your *.py or *.ipynb files using the designated folder on the K2 site of the course.

Project 1: Group 1 (Dioba Jordy COULIBALY, Anas BENELGHAZI, Luc GAULTIER), Group 2 (Pierre ACHARD, Enzo BEN HASSINE, Meygan CARDOSO FAGUNDES)

The task is to predict stock prices based on past stock prices. The data is stored in the file `tech_companies_stock_prices.csv`.

The fields are as follows:

- Date – the date of the transaction
- Stock_name – the name of the company
- Open, Close – opening and closing prices on the transaction date
- Open_{*i*}, Close_{*i*} – for $i = 1, 2, 3, 4, 5$, these represent the opening and closing prices on Date- i

For each company, predict the share price based on its past share prices over the previous k dates. Specifically:

- Try to predict both the opening and closing prices using various choices of past opening and closing prices.
- Explore using past share prices of other companies to predict the share price of a given company.

Use various models (such as linear regression, SVM, decision trees, and neural networks) to predict share prices. Compare the models in terms of prediction accuracy to determine which model performs best.

Finally, select one or two independent variables and use them to build models for predicting share prices. Identify which variables provide the best model for predicting share prices.

Project 2: Group 3 (Yixin Gao, Qianlei Weng, Xue Lu), Group 4 (Ngoc Quynh PHAM, Thanushree ANILKUMAR, Angella Maria SHAJU)

The task is to predict stock prices based on past stock prices. The data is stored in the file `stock_data_for_hw1.csv`.

The fields are as follows:

- Date – date of the transaction
- Company – name of the company
- Open, Close, High, Low – opening, closing, highest, and lowest prices on the transaction date
- Open- i , Close- i , High- i , Low- i – for $i = 1, 2, 3, 4, 5$, these represent the opening, closing, highest, and lowest prices on Date- i

For each company, predict the share price based on its past share prices over the previous k dates. Specifically:

- Try to predict opening, closing, high, and low prices separately, using different combinations of past opening, closing, high, and low prices.
- Investigate using past share prices of other companies to predict the share price of a given company.

Use various models (e.g., linear regression, SVM, decision trees, neural networks) for predicting share prices. Compare the models in terms of prediction accuracy to determine which one performs best.

Finally, select one or two independent variables to build models for predicting share prices. Identify which variables yield the best predictive model.

Repeat the exercise by replacing the logarithm of the price with the price itself (variable *price*).

Project 3: Group 5 (Hong Anh NGO, Paulina COTA, Simon RIEDL), Group 6 (HE Chengjie, ZHOU Haipeng, QU Siyi)

The task is to predict the logarithm of house prices (variable *lprice*) using the following variables:

- logarithm of the lot size (variable *llotsize*)
- logarithm of the surface area of the house (variable *lsqft*)
- number of bedrooms (variable *bdrms*)

- surface area of the house (variable *sqrft*)
- lot size (variable *lotsize*)

The data is stored in the file `hprice1_merged.csv`.

Use various models (such as linear regression, SVM, decision trees, and neural networks) to predict the logarithm of the house price. Compare the performance of the models in terms of prediction accuracy. Which model performs the best?

Select one or two independent variables and build models to predict the logarithm of the price. Identify which variables provide the best model for predicting the logarithm of the price.

Project 4: Group 7 (DAMI Tommaso, CASELLI Clara, FAUROUX Hannah), Group 8 (DULON Bettina, DAHAN Julie,Tang Pengfei)

The task is to predict the ESG score based on various other variables, such as:

- ESG Controversies Score
- Environmental Controversies Count
- Human Rights Controversies
- Business Ethics Controversies
- Wages Working Condition Controversies Count
- Total Renewable Energy To Energy Use in million
- Estimated CO2 Equivalents Emission Total

The data is stored in the file `ESGAllCompanies1.csv`.

Use various models (e.g., linear regression, SVM, decision trees, neural networks) to predict the ESG score. Compare the performance of the models in terms of prediction accuracy. Which model performs the best?

Select one or two independent variables and build models to predict the ESG score. Identify which variables provide the best predictive model for the ESG score.

Project 5: Group 9 (Fadi Ballegi, Jody Aria Widjaya, Rajneesh Rajbhar, Justin Durand)

Predict stock prices based on past stock prices, ESG Score, and net income. The CSV file `StockESG0.csv` contains the following columns:

- Date – date of transaction
- Company – name of the company
- Open, Close – opening and closing prices on the transaction date
- Open_{*i*}, Close_{*i*} – for $i = 1, 2, 3, 4, 5$, these represent the opening and closing prices on Date- i
- Net income – net income of the company
- ESG Score – overall ESG score
- ESG Controversies Score, Environmental Controversies Count, Human Rights Controversies, Business Ethics Controversies, Wages Working Condition Controversies Count, Total Renewable Energy To Energy Use in million, Estimated CO2 Equivalents Emission Total – various ESG-related scores and metrics

Use past opening or closing prices, net income, and various ESG scores to predict the current opening (or closing) price.

Apply different models, such as linear regression, SVM, decision trees, and neural networks, for predicting the share price. Compare the models in terms of prediction accuracy to determine which performs best.

Finally, select one or two independent variables to build predictive models for share prices. Identify which variables yield the most accurate model.